Anay Basu

anayb1490@gmail.com | www.linkedin.com/in/anay-basu/ | www.github.com/anayb1490

SUMMARY

Data and AI engineer with a Computer Science and Behavioral Neuroscience degree who builds data pipelines and LLM systems end to end in Python and SQL. Built and benchmarked LLM methods on the OpenAI API, including RAG, built Python data pipelines for research analysis, and built a full-stack generative AI application.

EDUCATION

Northeastern University *August 2025*

Bachelor of Science in Computer Science and Behavioral Neuroscience

EXPERIENCE

Bond Health, Inc.

AI Research Consultant *June 2026 – August 2026*

- Built an LLM system on the OpenAI API that matches patients to clinical trials by judging eligibility criterion by criterion, evaluating a range of methods including retrieval-augmented generation (RAG)
- Benchmarked five approaches against physician labels, with the best reaching 77% accuracy
- Diagnosed a systematic bias in the model's overall verdicts and corrected it by deriving them from its own criterion-level outputs, lifting trial-level accuracy from 57% to 64%

Codvo.ai

Data Science Intern *September 2025 – December 2025*

- Validated performance of client machine learning models in Python (NumPy, Pandas, Scikit-learn), analyzing accuracy and reliability metrics to assess model quality

Northeastern University

Data Science Intern *July 2024 - August 2024*

- Built Python (Pandas, NumPy) data pipelines to analyze household-level spillover effects of a large-scale environmental program across 1,500+ households, with findings included in a research report
- Visualized variable impacts across treatment groups to communicate findings to academic and policy stakeholders

Beth Israel Deaconess Medical Center

Neuroimaging Research Assistant *July 2023 - December 2023*

- Trained and compared CNNs and other deep learning architectures in PyTorch for cerebral microbleed segmentation in the Soman Lab, building DICOM preprocessing pipelines for model input
- Reduced manual clinician review time by 30% by automating microbleed detection in neuroimaging scans

PROJECTS

Ariadne (Full-Stack + Agentic AI System) *April 2025 – Present*

- Built a screen-aware generative AI tutor (Python, FastAPI, Gemini API, Supabase, Electron) that uses tool calling to decide when to capture the student's screen and tailors help to what each assignment is meant to teach, with a real-time voice mode over the Gemini Live API
- Designed a privacy-preserving instructor dashboard that clusters class questions with an LLM, enforcing anonymity thresholds and server-side counts in code rather than trusting the model

Unblockable (Android App) *May 2026 – Present*

- Built a native Android app in Kotlin using Jetpack Compose that intercepts designated app launches via UsageStatsManager and renders a full-screen cognitive puzzle overlay (Stroop, pattern, spatial, logic) before granting access, converting reflexive doomscrolling into a deliberate choice
- Designed procedural puzzle engine, foreground detection service with battery optimization handling across OEM Android skins, and a Room-backed usage dashboard tracking daily and weekly attempt patterns, currently in closed beta via Google Play

SKILLS

- **Languages:** Python, SQL, R, TypeScript, JavaScript, Java, Kotlin
- **Frameworks & Tools:** FastAPI, Supabase, Electron, Git, Docker, AWS, Jetpack Compose
- **AI/ML:** LLMs (OpenAI API, Gemini API, tool calling), Retrieval-Augmented Generation (RAG), Prompt Engineering, NLP, Model Evaluation, PyTorch, TensorFlow, Scikit-learn, Deep Learning
- **Data & Analysis:** Pandas, NumPy, Data Pipelines, Data Analysis and Reporting, Statistical Analysis, Data Visualization, Matplotlib, Time Series Analysis
- **Domain Knowledge:** Neuroscience, Neuroimaging, Healthcare Analytics, Cognitive Science, Biology, Medical Imaging